



RoCam.

High Performance Vision-Guided Rocket Tracker

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Web UI & UX Design

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Arch. Backend & Firmware

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Data Tooling & i18n

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CV, ML & System Tests

TEAM #3 SPACEY | MCMaster UNIVERSITY SFWRENG 4G06

THE CHALLENGE

Rockets reach **Mach 3** in seconds.

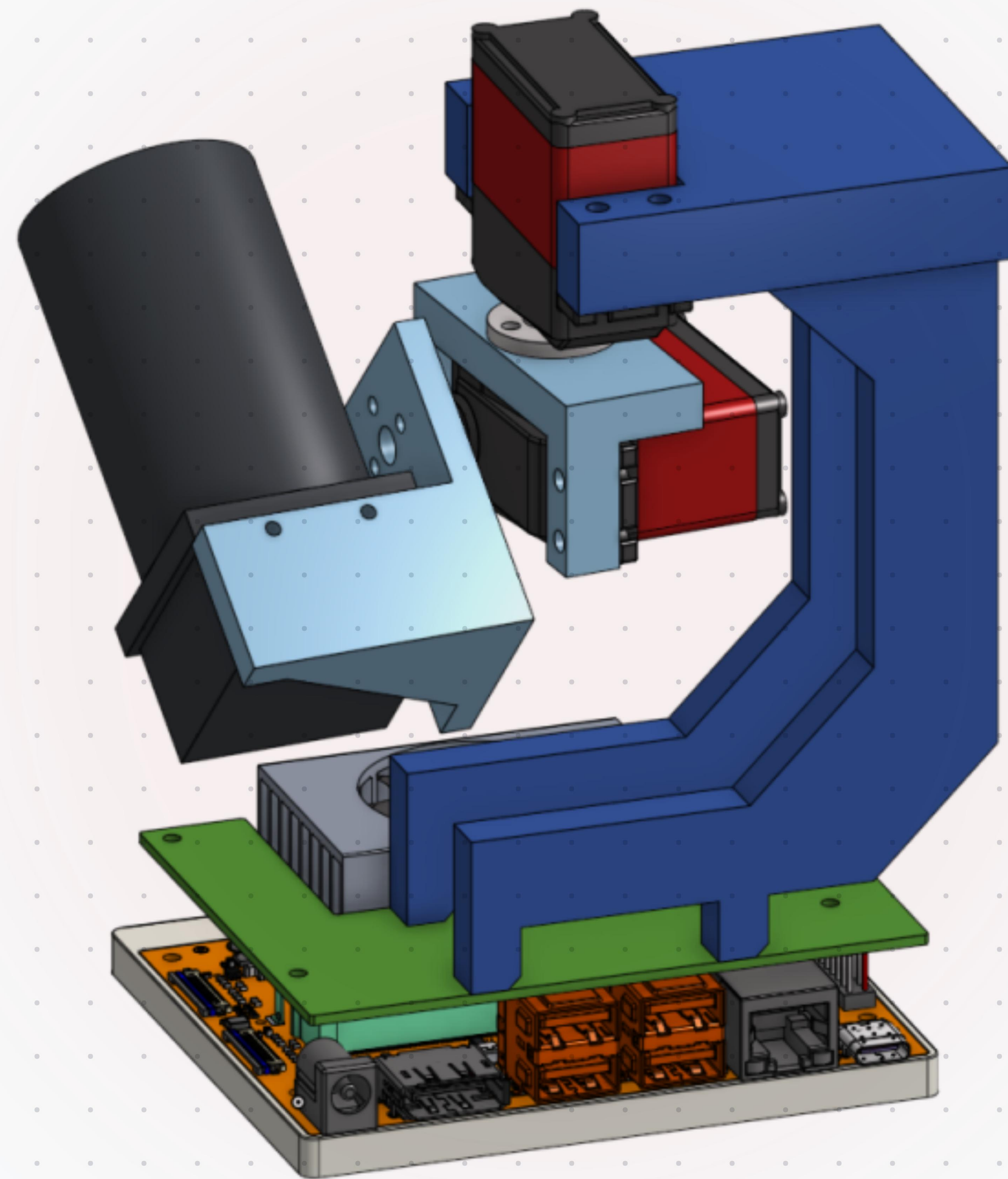
Manual optical tracking is physically impossible for small-profile hobbyist rocketry. Commercial rigs cost upwards of \$10,000.

100 km
MAX ALTITUDE

<1 s
EVENT DURATION

PRESENTATION ROADMAP

- 1. Problem & Solution
- 2. Architecture & Live Demo
- 3. Data, CV & ML Pipeline
- 4. Hardware & UI Control
- 5. Testing, Challenges & Future



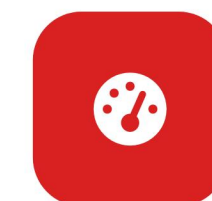
Autonomous. Accessible. Accurate.

A fully autonomous, closed-loop optical tracking system engineered to meet strict capstone constraints.



Edge AI Tracking

YOLO + TensorRT on Jetson Orin Nano



High-Speed Loop

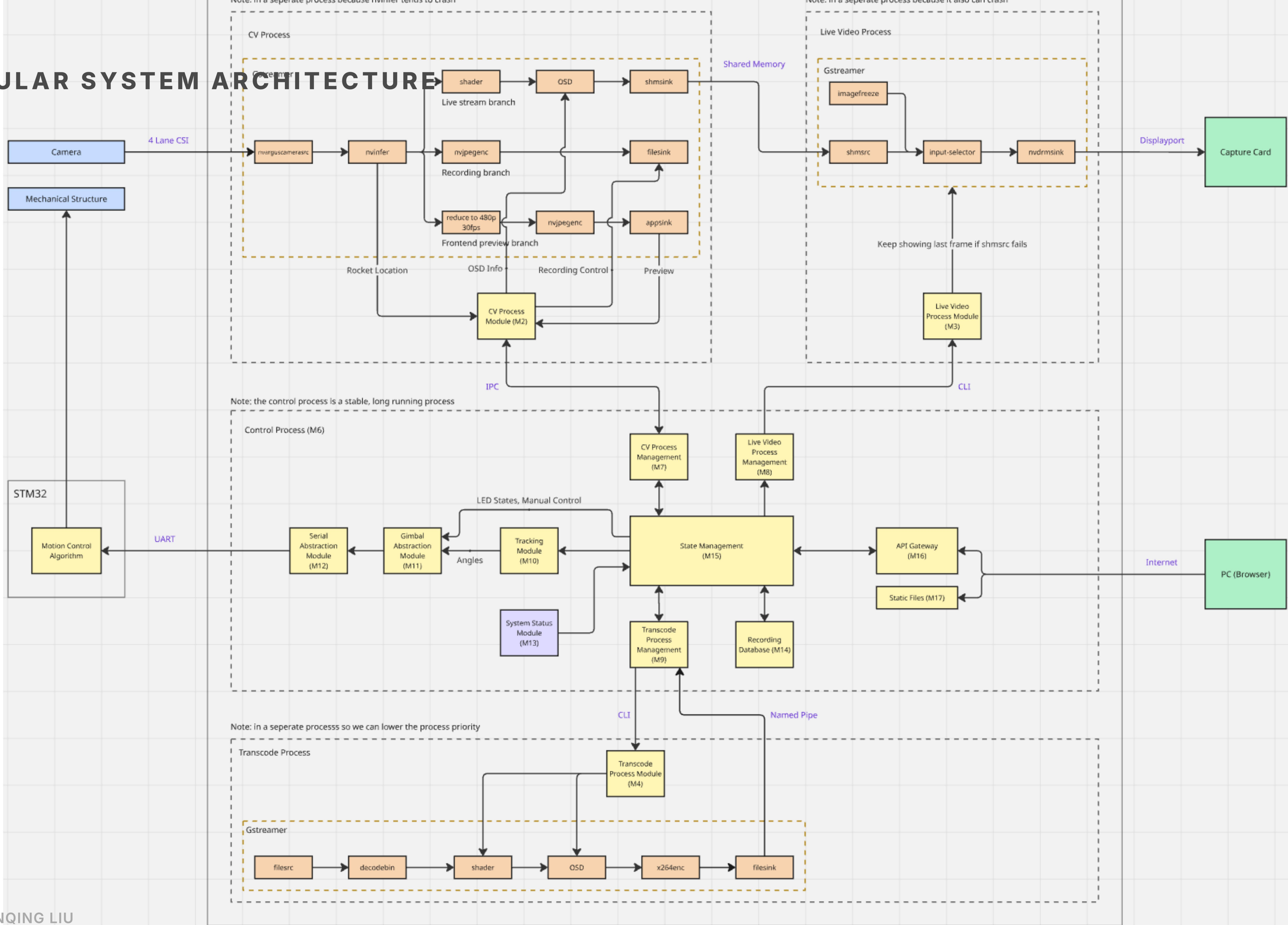
60Hz closed-loop servo actuation

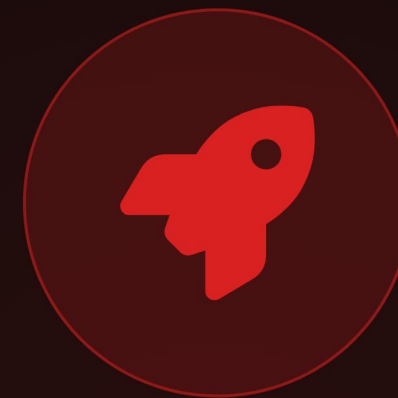


Budget Constraint Met

Total BOM under \$500 CAD (vs \$10k+ commercial)

MODULAR SYSTEM ARCHITECTURE





LIVE DEMO

AUTONOMOUS TRACKING IN ACTION

Custom labelling platform for rocket data.

All training data flows through our in-house **RoCam Labeler** — a full-stack annotation tool built from scratch to handle our unique dataset requirements.



SAM-Assisted Annotation

One-click segmentation masks via Meta SAM



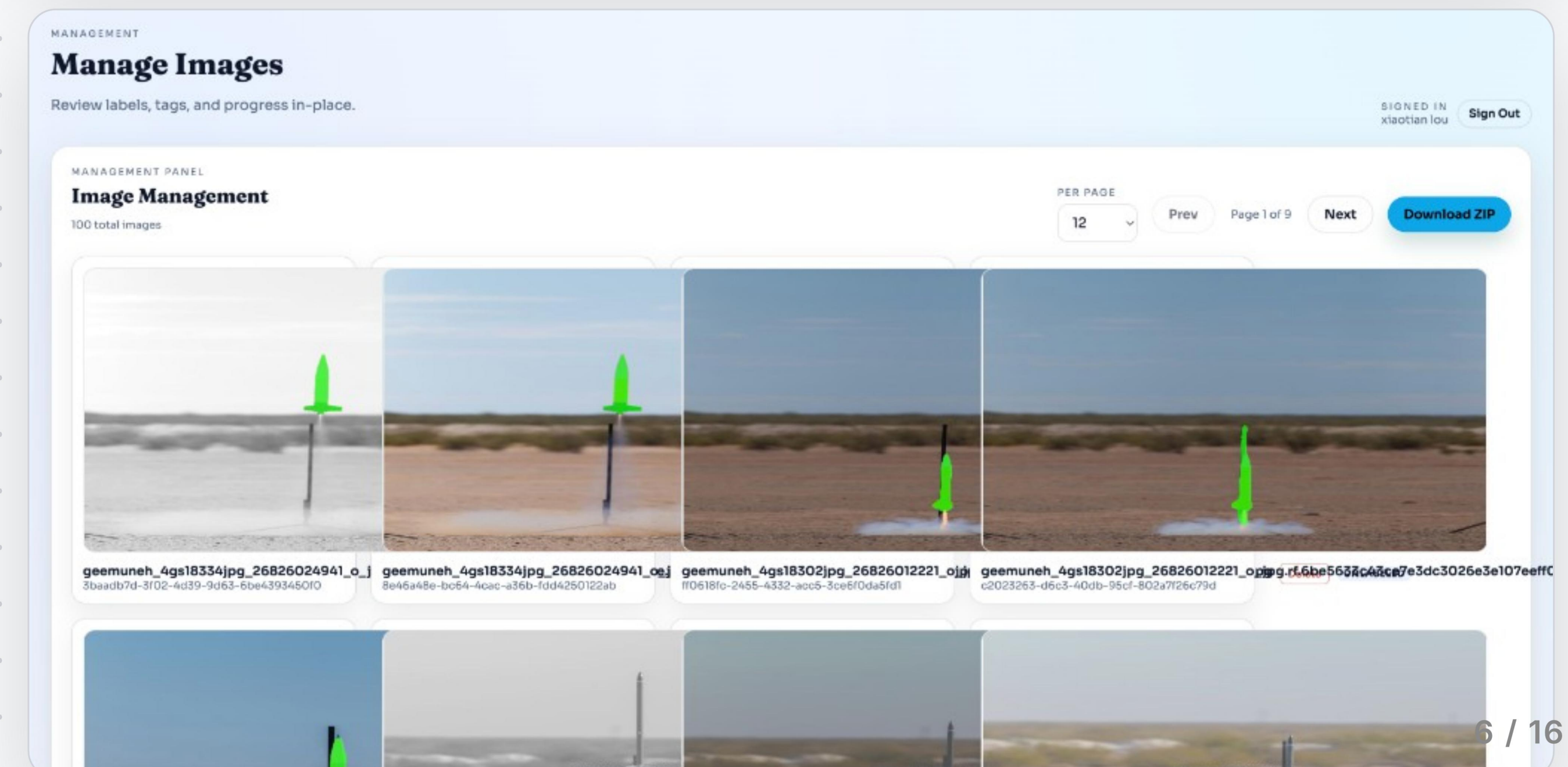
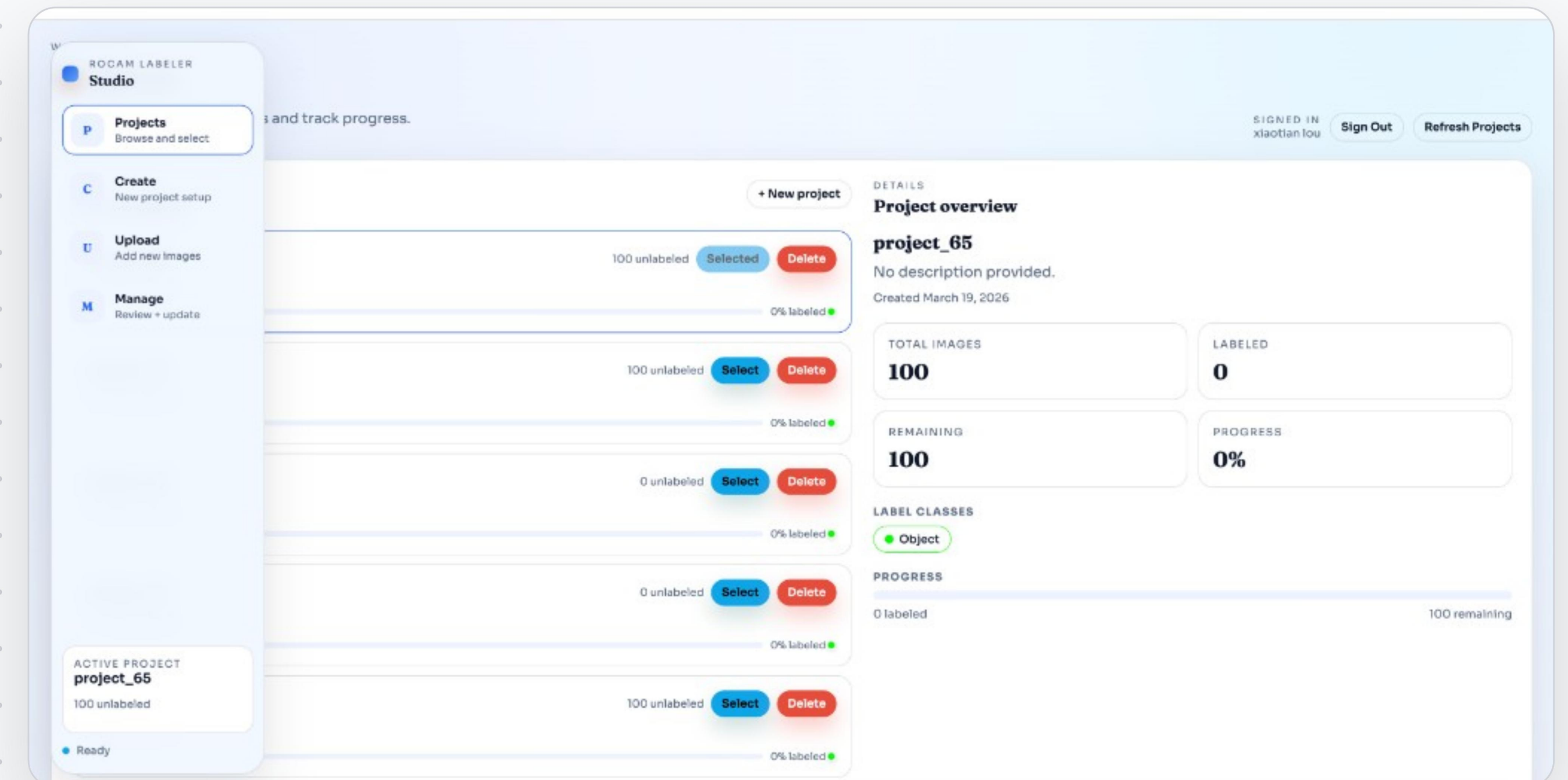
Project Management

Multi-project workspace with progress tracking

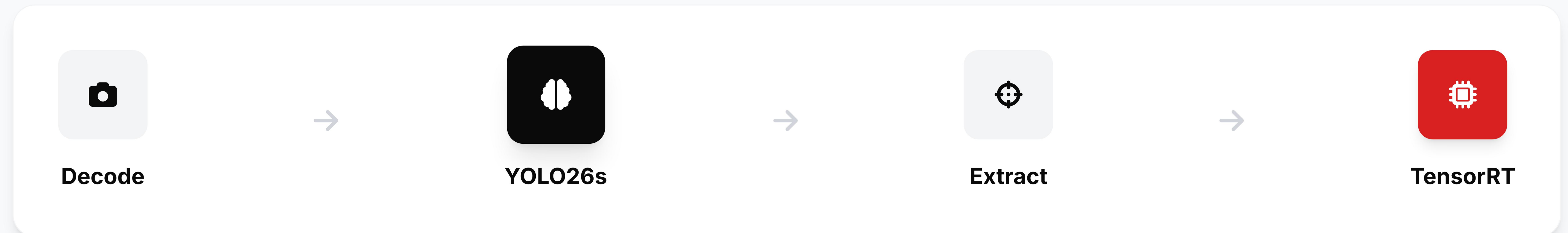


Batch Upload / Export

ZIP-based bulk import and YOLO-format export



Optimized for extreme distance and strict latency budgets.



🔍 Why YOLO26s-P2?

YOLO26s with a custom P2 stride-4 detection head excels at micro-objects. Critical when the rocket occupies <1% of the frame at extreme range.

🔴 NVIDIA DeepStream

Hardware-accelerated pipeline ensures that frame decoding and TensorRT FP16 inference happen strictly under our 16.6ms per frame budget.

0.958

FINAL MAP@50 SCORE

+22%

MAP@50-95 IMPROVEMENT

Over baseline COCO weights during extreme weather validation.

3-Stage Training Pipeline

Training data labelled via our custom **Labelling App** (built by Mike Chen) with SAM-assisted annotation.

1

Stage 1: Full Training (300 epochs)

COCO pre-trained YOLO26s-P2 fine-tuned on rocket dataset with mosaic augmentation (0.4) and zero patience.

2

Stage 2: Rect Fine-tune

Resume with rect=True and hard-negative backgrounds (empty skies, birds) to suppress false positives.

3

Stage 3: Low-LR Polish

Final epochs at reduced learning rate on curated, manually verified subset for maximum precision.

Bare-metal **Rust** for hard real-time execution.

60 Hz

CONTROL LOOP

Embassy

RUST ASYNC

CRC-8

DATA INTEGRITY

<0.5s

WATCHDOG TRIP



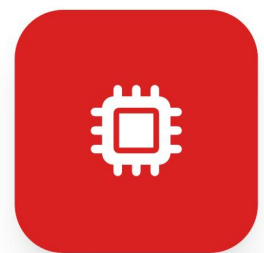
Jetson Orin

Vision / Math

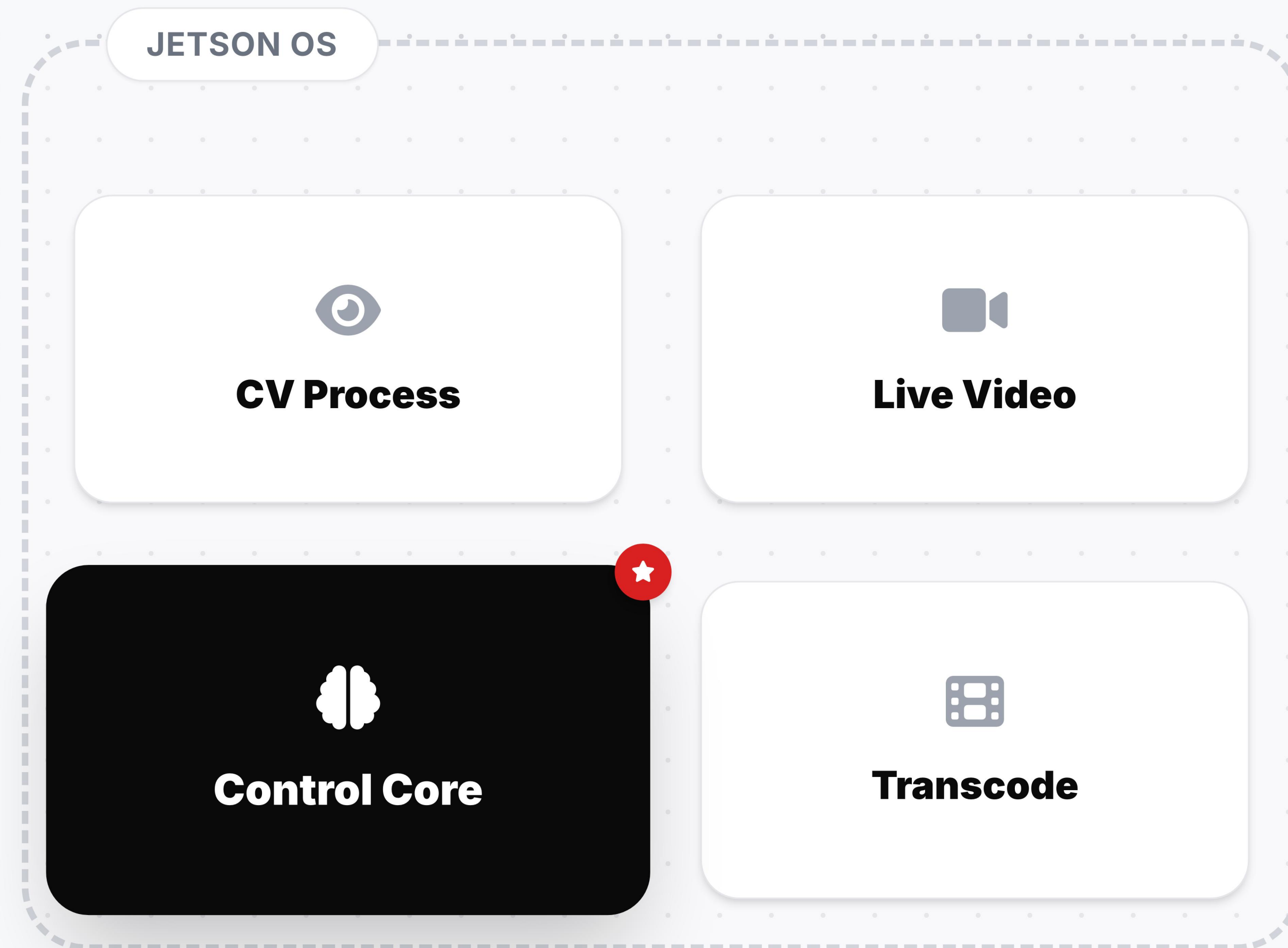
115200 BAUD UART

STM32F405

Custom PCB



FAULT-TOLERANT ARCHITECTURE



Built for isolation and real-time survival.



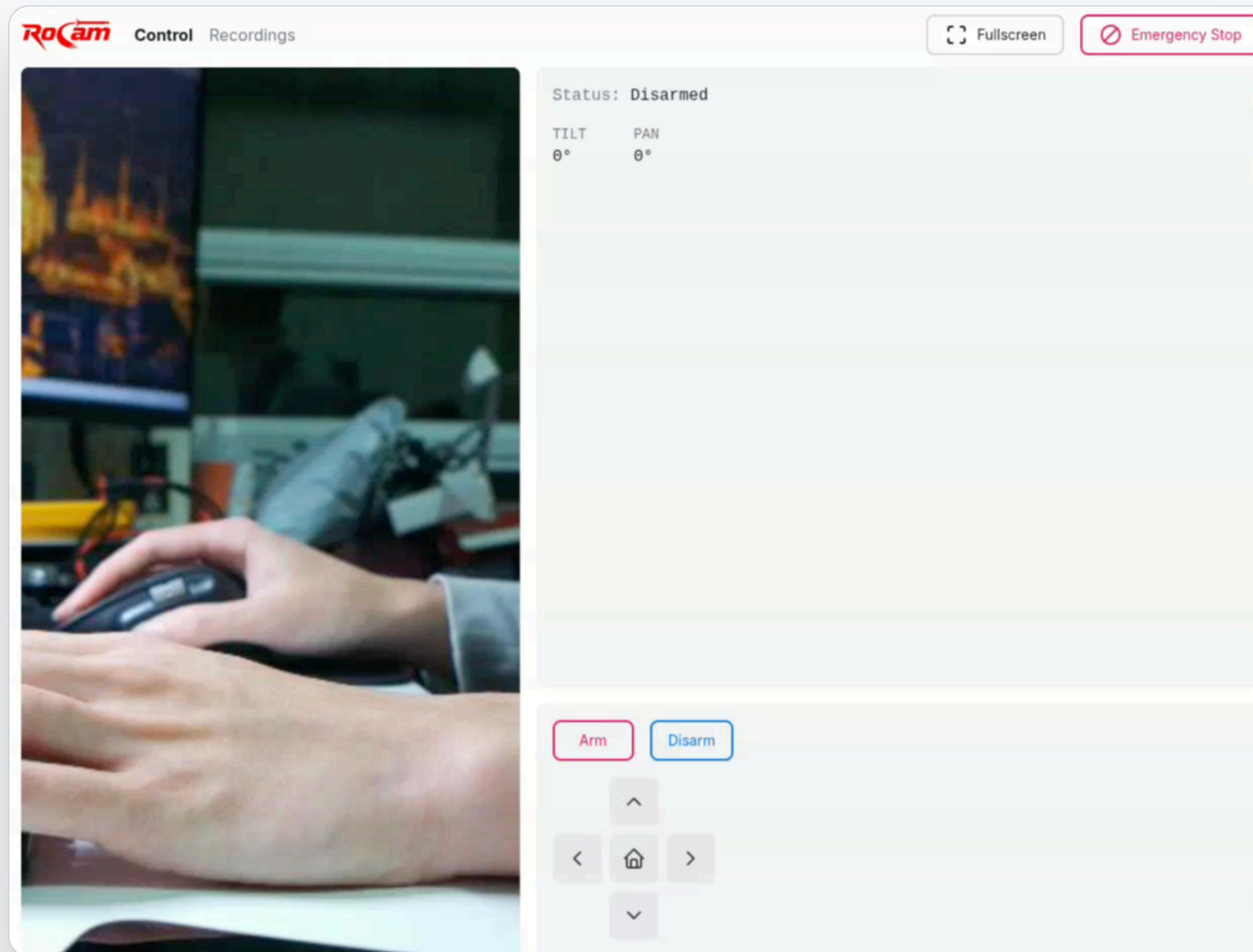
Independent Restart

If the CV inference crashes, the control process maintains connection to the gimbal, preventing lock-up.



Zero-copy IPC

Shared-memory architecture allows massive 1080p frames to move between processes instantly.



Command center in a browser.

REACT • TYPESCRIPT • VITE

🎥 Low-latency live preview

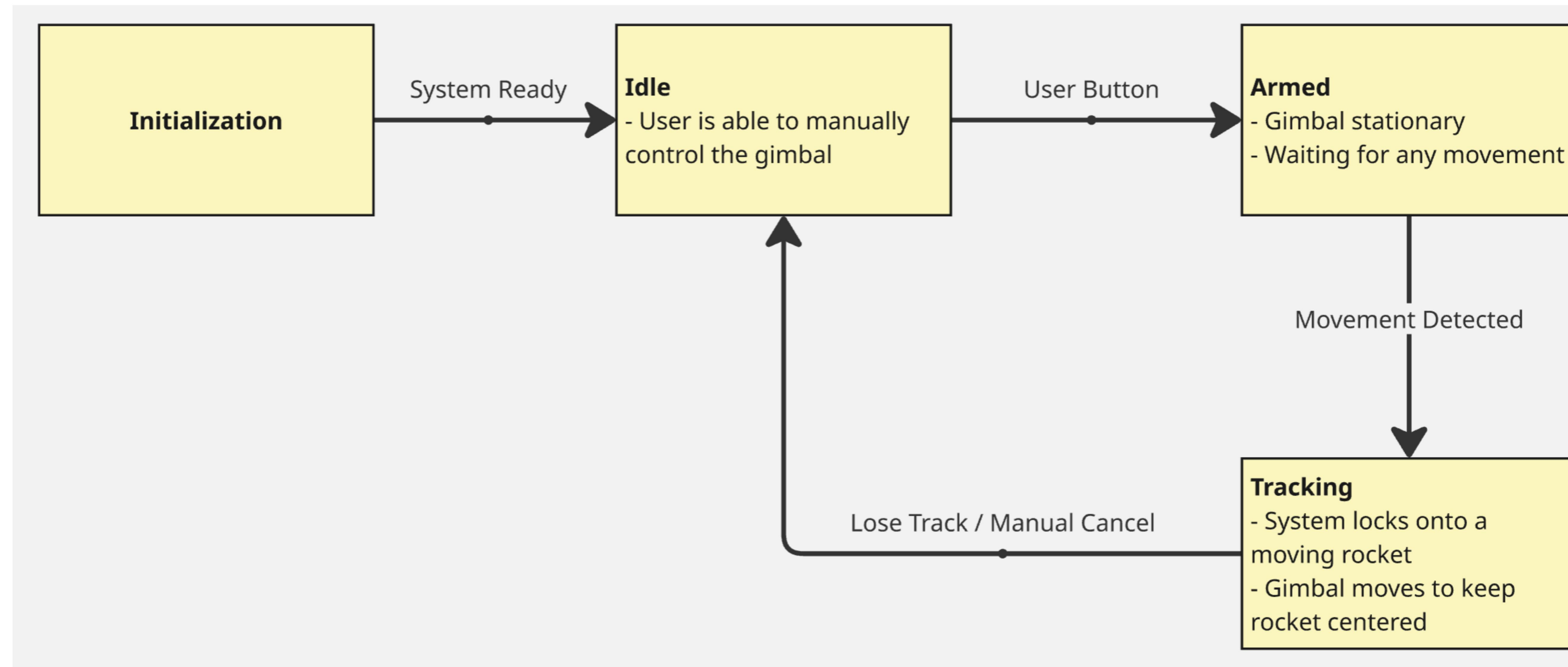
⚙️ Direct manual gimbal override

📈 Real-time health telemetry

🌐 Full Bilingual (EN/FR) support

STATE MACHINE & SAFETY

⚠ Safety: Hardware Limits Enforced



Disarmed

Safe mode. Motors off, manual UI control only.

Armed

Motors active. CV watching for launch event.

Tracking

Rocket detected. Closed-loop PID taking over.

Recording

Event complete. Saving H.264 footage to disk.

481

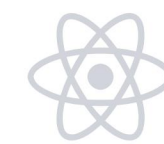
AUTOMATED TESTS PASSED



Backend (pytest)

313 Unit Tests

88% Coverage



Frontend (Vitest)

168 Component Tests

86% Coverage



System E2E

GitHub Actions CI

100% Pass Rate

— PROJECT CHALLENGES

Real-time Latency

Python's GIL blocking CV loops.



Solution

Shared-memory IPC bypassing GIL entirely.

Micro-Object Detection

Rockets vanishing into background noise.



Solution

YOLO26s-P2 Head & Hard-negative mining.

Hardware Sync

UART dropping packets at 115200 baud.



Solution

Logic analyzer debug + custom CRC-8 checksums.

Designed to scale from 200m hobby rockets to 3km+ high-power launches.



Performance

- 4K @ 60fps tracking
- TensorRT INT8 Quantization
- Optical Auto-focus



Hardware Scaling

- Full-size Gimbal Mounts
- Multi-camera Arrays
- Weatherproof Enclosures

NEXT STEP



Software UX

- Mobile-native Apps
- Post-flight Analytics
- AR Telemetry Overlay



Thank You.

Questions?



github.com/ZifanSi/vision-guided-tracker

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